SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

ARTICLE REVIEW

Code of Conduct:

I Priyanka.G / 192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Article Selected

Topic Area: Decision Trees and Statistical Learning Methods (Logistic Regression), benchmarked against each other on a classification task

Belinda, C. M. J., et al. "Evaluating Logistic Regression, Decision Tree, and Random Forest for Predicting Stroke Risk." Proceedings of the 2025 2nd Asia Pacific Conference on Innovation in Technology (APCIT), IEEE, 2025.

IEEE Xplore Link: https://ieeexplore.ieee.org/document/11411440

Publisher: IEEE Xplore 2025 2nd Asia Pacific Conference on Innovation in Technology (APCIT)

Task 1: Article Summary

(a) Citation, Domain, and the Problem Being Solved

Citation: Belinda, C. M. J., et al. "Evaluating Logistic Regression, Decision Tree, and Random Forest for Predicting Stroke Risk." 2025 2nd Asia Pacific Conference on Innovation in Technology (APCIT), IEEE Xplore, 2025.

Domain: Clinical decision support specifically, early screening for stroke risk from routinely collected patient data.

The ML Problem: This is a binary classification task in the strictest sense: predict stroke versus no stroke from a patient's demographic and clinical profile. What makes it worth reading isn't the task itself, which is a familiar one in health informatics, but the choice to run three quite different learners' side by side on the same pipeline.

(b) What the Paper Is Actually Doing

The motivation is not exotic: stroke prediction has been studied many times over, and the authors are upfront that their contribution is comparative rather than novel. Their real target is a gap that shows up repeatedly in this literature studies that report a single model's accuracy without addressing the fact that stroke datasets are heavily skewed toward non stroke cases, so a model can look accurate while quietly failing on the patients who matter most. Rather than proposing a new architecture, the authors take three familiar algorithms Logistic Regression, Decision Tree and Random Forest and put them through an identical preprocessing pipeline: missing value imputation, categorical encoding, feature scaling, and SMOTE oversampling to rebalance the minority stroke class. Each model is then trained on a stratified split and scored on accuracy, precision, recall, and F1. The value of the paper lies less in any single number and more in showing, side by side, how differently a linear statistical model and a tree based inductive learner respond to the same imbalanced dataset once that imbalance is corrected for.

Task 2: Methodology Analysis

(a) Technique and Dataset

Three learners appear on the paper, and each map onto a different corner. Logistic Regression is the statistical baseline: a linear model estimating the log odds of stroke from the input features, valued here for being fast, interpretable and easy to sanity check. The Decision Tree is the paper's clearest example of inductive learning a CART style classifier that recursively partitions the feature space using an impurity measure such as Gini index, producing a set of if then rules a clinician could actually read. Random Forest sits on top of the same idea, training many such trees on bootstrapped samples and averaging their votes to trade away some interpretability for a meaningful gain in stability.

The dataset is a publicly available stroke risk collection with patient level demographic fields (age, gender), lifestyle indicators (smoking status, occupation), and clinical measurements (hypertension, prior heart disease, average glucose level, BMI), tied to a binary stroke label. Preparation followed a fairly standard route: missing BMI values were imputed, categorical fields such as gender and work type were label encoded, continuous features were scaled, and SMOTE was used to synthetically oversample the minority stroke class before the stratified train test split. None of this is unusual, which is part of why the comparison is a fair one the differences in outcome can reasonably be attributed to the algorithms rather than to inconsistent preprocessing across the three runs.

(b) Mapping, and Where the Method Holds Up (or doesn't)

The Decision Tree is the piece that speaks most directly to inductive learning and decision tree strands, since its splits are derived purely from the training data rather than assumed in advance. Logistic Regression represents the statistical learning side of the same outcome, and Random Forest is best read as an extension of the decision tree idea into an ensemble setting rather than a separate technique in its own right.

What the authors got right: Applying SMOTE before training is the correct instinct for a dataset this imbalanced, and it's the reason the Decision Tree's recall figure is worth trusting at all without it, a tree trained on the raw class ratio would likely have learned to predict “no stroke” almost by default and still posted a deceptively high accuracy.

Where I'd push back: The methodology section does not make it fully clear whether SMOTE was applied before or after the train test split. If it was applied beforehand, synthetic points derived from the same underlying minority samples could end up on both sides of the split, which would inflate every reported metric, not just accuracy. This is a common and easy mistake to make, and it's the first thing I would ask the authors to clarify.

Task 3: Results & Critical Evaluation

(a) What the Numbers Say

Random Forest comes ahead on test accuracy, with the Decision Tree landing between Logistic Regression trailing both models. In the paper’s discussion, the Decision Tree’s performance is described as particularly balance balanced its ion and recall are relatively close to each other suggesting it makes fewer extreme tradeoffs (for example, avoiding a situation where it only catches stroke cases at the expense of many false alarms, or vice versa). The authors also use the confusion matrix to strengthen this interpretation: they report that Random Forest produces the fewest overall misclassifications, meaning it reduces both types of errors false positives (flagging patients who do not have a stroke) and false negatives (missing patients who do). Conceptually, this pattern makes sense because Random Forest aggregates predictions from many bootstrapped decision trees, which tends to smooth out the weaknesses of any single tree (such as sensitivity to particular split choices or local noise in the training data). In contrast, a single Decision Tree may capture useful decision rules but can be more vulnerable to variations in the training set, which can limit its consistency on unseen test data. Logistic Regression, while still a valuable baseline due to its simplicity and interpretability, is constrained by its linear decision boundary assumptions, so it may struggle to fully represent complex relationships among features when stroke risk depends on nonlinear interactions between variables. Overall, the table is included to allow a direct comparison of these model's side by side, so the reader can connect performance differences not only to headline accuracy, but also to the underlying error patterns reflected in precision, recall, and the confusion matrix.

Model	Test Accuracy	Reported Behavior	My Read
Logistic Regression	79.18%	Retained mainly for its interpretability, not its accuracy	Useful floor, not a serious contender
Decision Tree	89.46%	Noticeably better sensitivity; precision and recall stay close together	The most instructive result for
Random Forest	95.01%	Fewest misclassifications on both sides of the confusion matrix	Best performer, at some cost to transparency

(b) A Closer Look at Whether These Results Hold Up

Taken at face value, none of these numbers look implausible they sit comfortably within the range other stroke prediction studies on similar open datasets have reported, so there's no obvious red flag of overfitting or metric manipulation. That said, I'd want to see three things before fully trusting the headline accuracy figures. First, as noted above, whether the SMOTE resampling happened before or after the split matters a great deal here, and the paper doesn't settle the question. Second, accuracy is a blunt instrument on a dataset this imbalanced; what actually matters clinically is the recall on the stroke positive class specifically, and that figure is not broken out separately in the results the paper reports. Third, the dataset comes from a single open repository, which raises the usual concern about whether the patient population is representative of the settings where this model would eventually be deployed different hospitals, different regions, different diagnostic conventions could all shift these numbers considerably. A k fold cross validation run, together with an external validation set drawn from a different source, would go a long way toward settling whether these results are robust or an artefact of one particular data split.

(c) Where This Could Actually Be Used

The most realistic deployment for a model like this is not as a diagnostic tool but as a triage layer something that sits inside an electronic health record system and flags patients whose profile warrants a closer look, rather than something that makes a final call on its own. The Decision Tree is arguably the more deployable of the three models precisely because its rules are auditable; a clinician can trace exactly why a given patient was flagged, which matters both for trust and for regulatory sign off. Scaling this beyond a single study is where the harder problems start hospital data quality and coding conventions vary widely, the model would need periodic recalibration as patient populations shift, and there are real ethical stakes in a system that can produce false negatives for a life-threatening condition. None of these are reasons to dismiss the approach, but they are reasons the paper's promising numbers shouldn't be read as deployment ready as they stand.

Task 4: Reflection & Learning Outcome

(a) What This Article Actually Clarified for Me

Reading the Decision Tree results alongside the raw splitting logic made the idea of inductive learning feel far less abstract than it does on the whiteboard watching a tree pull out age, glucose level and BMI as its early splits was a much more concrete way to understand "learning rules from labelled data" than any definition I'd been given in class.

The second thing that stuck with me is how easy it would have been to dismiss Logistic Regression outright given its lower accuracy, and how wrong that instinct would have been. Its role here as an interpretable baseline is exactly the point about statistical learning methods: they're not there to win; they're there to give you something honest to measure the more complex models against.

The third insight was less about any one algorithm and more about the order of operations. Seeing how much the class imbalance correction shaped every downstream result drove home that, in terms, the choice of model and the choice of how you prepare the data are not separate decisions get the imbalance handling wrong, and it barely matters which classifier you picked afterward.

(b) An Extension I'd Actually Want to Run

If I were taking this further, I would keep the same three model comparison but change two things: swap the single hospital open dataset for a federated setup spanning several institutions and add an explainability layer SHAP values would be the obvious choice on top of the Decision Tree and Random Forest outputs, paired with a cost sensitive training objective that penalises false negatives more heavily than false positives.

None of these requires new infrastructure. Cost sensitive variants of scikit learns tree-based classifiers and SHAP explainability tooling are both freely available, and federated learning frameworks such as Flower or TensorFlow Federated already support training across institutions without centralising patient records, which sidesteps a good part of the data access problem raised in Task 3(c).

The expected payoff is fairly specific: better recall on the stroke positive class, since that's what a cost sensitive objective directly targets; more clinician trust, since SHAP attributions give a concrete reason for each flagged case rather than a bare probability; and a model that generalises past the quirks of one dataset's demographic makeup, which is the exact limitation flagged in Task 2(b).